GPU-powered AI in docker-based Jupyterlab on Galaxy EU

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An open-source, docker-based, and GPU-enabled Jupyterlab [1] notebook has been developed running on the public compute infrastructure of Galaxy Europe for rapid prototyping and developing end-to-end artificial intelligence (AI) projects. Using this Jupyterlab notebook, long-running training jobs can be executed remotely on Galaxy's infrastructure.

Docker-based Jupyterlab

- Shreable AI model using ONNX format
- Git integration
- GPU enabled using CUDA (Nvidia)
- Send long-running AI training jobs to Galaxy cluster
- Workflow of notebooks using Elyra AI
- Large number of data science packages
- Docker image [5] containing all necessary packages

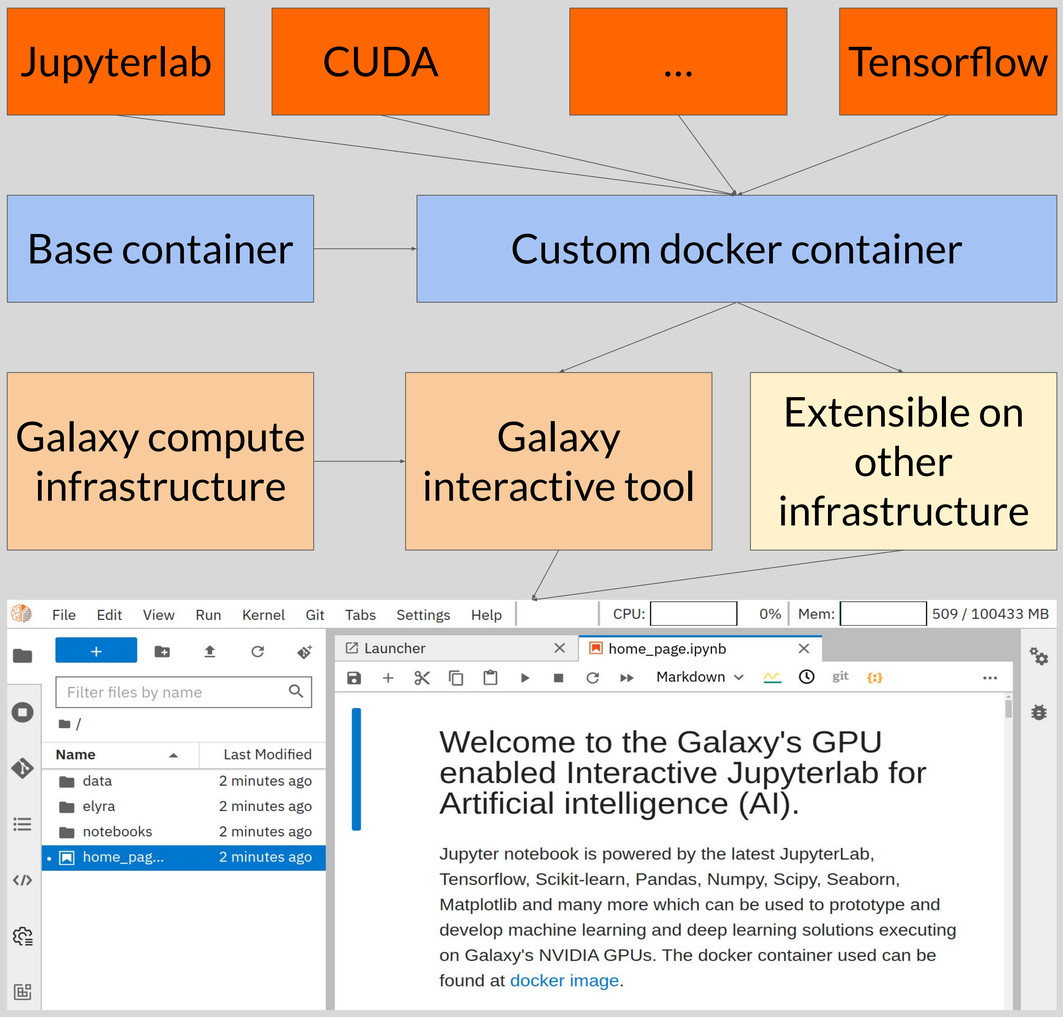

Figure 1. Architecture of Docker-based Jupyterlab

Galaxy's Interactive tool

- Pull above docker image and open Jupyterlab on Galaxy
- Run on GPU, large compute resources i.e. RAM, disk space
- Access Galaxy's history, datasets using Bioblend
- Accessible via Galaxy Europe as a tool
- Scan QR code (bottom) to access its tutorial on GTN [6]

Scientific analyses (use-cases)

- **COVID CT Scan image segmentation** (Figure 2) [2]
  - Deep learning architecture using Unet
  - Input data as H5 files
  - Remote training job on Galaxy cluster
  - Trained models in Galaxy history as ONNX file
  - Download trained model and make inference

- **3D structures of proteins using Colabfold** (Figure3) [3]
  - Alphafold2 [4] weights for prediction
  - 3D structure using only protein sequence
  - Faster prediction on GPUs in Jupyterlab notebook
  - Download structure predictions for further analyses

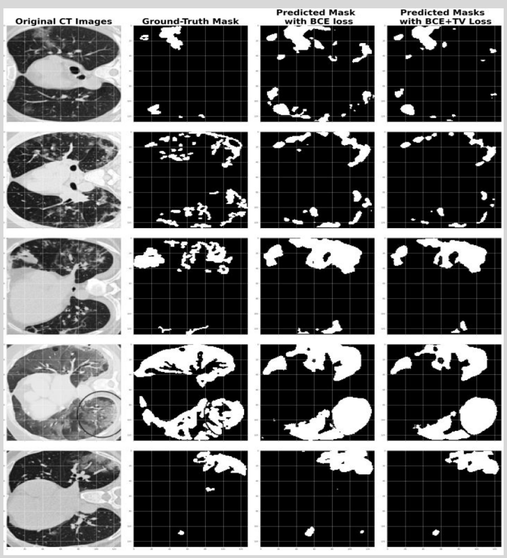

Figure 2. Ground truth and predicted masks using Unet deep learning architecture

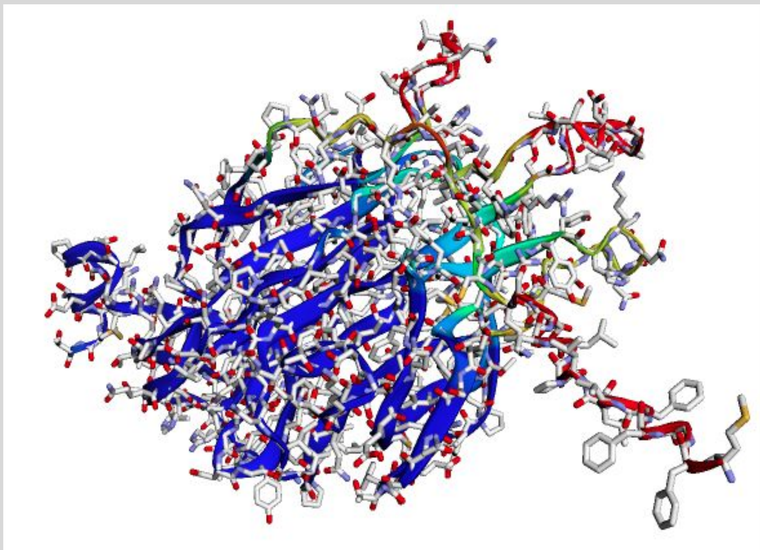

Figure 3. 3D structure of a protein sequence predicted using Colabfold

References

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5. Docker image: anupkumar/docker-ml-jupyterlab:galaxy-integration-0.1. Github: https://github.com/anuprulez/ml-jupyter-notebook/tree/galaxy-inte-docker-ml-jupyter
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